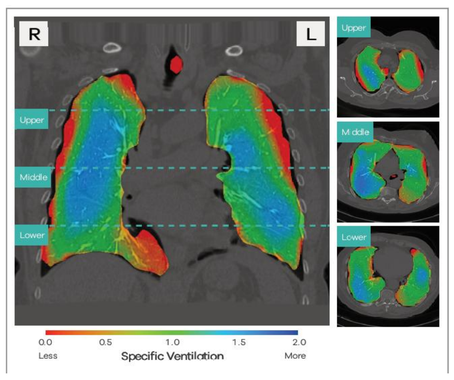
# LungLogic: Intelligent Airflow Control for Realistic Breathing Simulation

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## 1. Introduction

### 1.1 Project need

Dynamic lung phantoms provide repeatable breathing inputs for imaging and ventilation research.

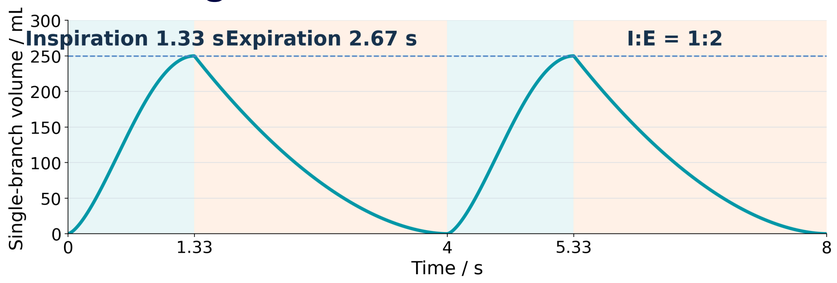



### 1.2 Project objective

LungLogic controls airflow in a biomimetic lung model. This stage validates one 250 mL branch before bilateral integration.

Regional ventilation imaging. Application context.

### 1.3 Nominal breathing reference




*Breathing-cycle illustration at 15 bpm and I:E = 1:2.*

| | |
|---|---|
| **RR · respiratory rate** | 12–16 bpm; nominal 15 bpm |
| **I:E · inhale:exhale time** | 1:2; 1.33 s / 2.67 s at 15 bpm |
| **VT · tidal volume** | 250 mL per branch |
| **Repeatability** | VT error within ±5% |

*Nominal design values, not measured performance. Target: 250 mL per branch and VT error within ±5%.*

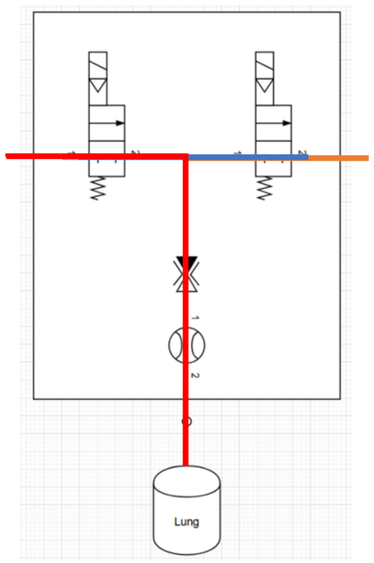
Sources: 4DMedical project scope (19 March 2026), pp. 5, 13. Ventilation image: supplied reference.

## 2. Pneumatic System & Hardware

### 2.1 Objective

Control one lung branch and measure airflow for closed-loop regulation and calibration.

### 2.2 Pneumatic system prototype




**Prototype photograph (to be added)**

*Fig. 1. Single-lung route; prototype photograph pending.*

### 2.3 Embedded control architecture

**PC / Python UI** → **Controller** → **24 V bridge PCB** → **Valves + sensor**

The controller runs the 100 Hz breathing and flow loop. The bridge connects logic to the routing valves, PPV and SFM3300 sensor.



*Fig. 2. Control architecture and bridge PCB.*

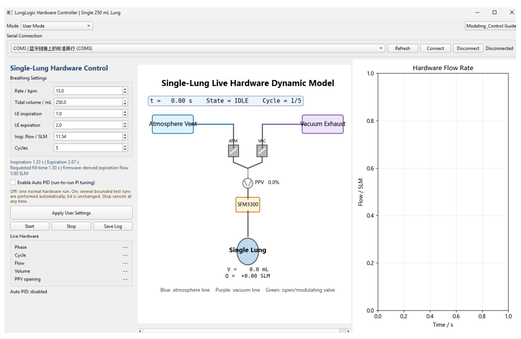
## 3. Integrated Control System & Software

### 3.1 Objective

Track flow and tidal volume with a visible, repeatable closed-loop system. Feedforward supplies the nominal valve command; sensor feedback corrects error.

### 3.2 Operator and developer interface

**User Mode**




**Developer Mode**

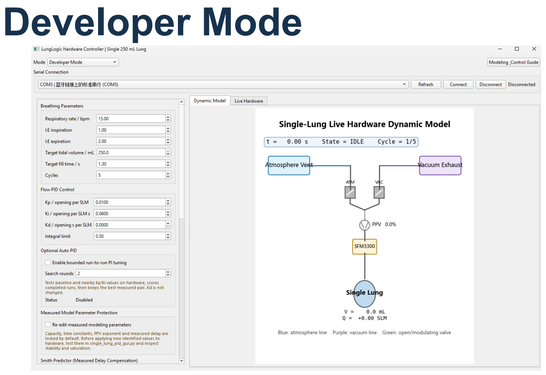
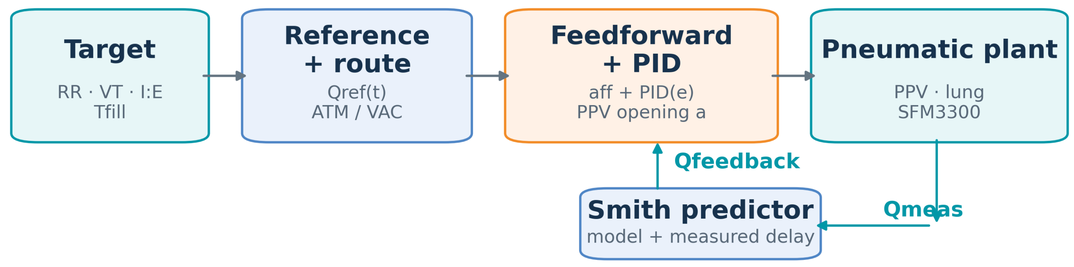



*Fig. 3. User and Developer modes.*

### 3.3 Model-assisted flow feedback




$$Q_{ref,\,in} = 0.06 V_T / T_{fill} \quad \cdot \quad e = Q_{ref} - |Q_{feedback}| \quad \cdot \quad a = \mathrm{clip}(a_{ff} + PID(e), 0, 1)$$

*Fig. 4. Feedback control with optional Smith prediction.*

### 3.4 Auto PID and measured-delay compensation

Auto PID scores bounded Kp/Ki trials. Smith prediction uses the identified model and measured delay; hardware adds no artificial delay.

## 4. Results & Future Work

### 4.1 Experimental results

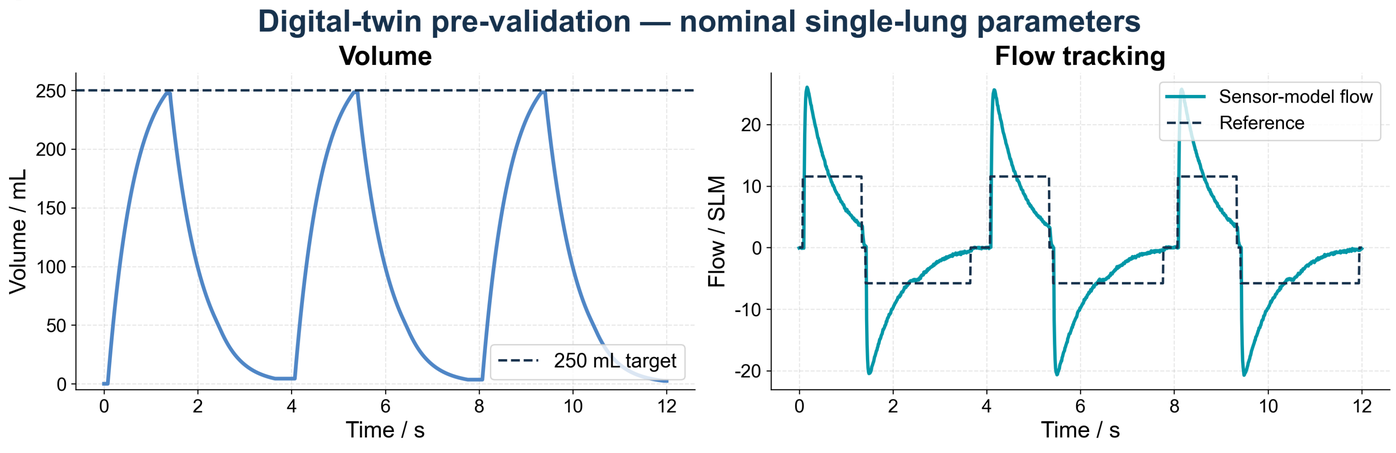



*Fig. 5. Digital-twin simulation at 15 bpm (not measured hardware).*

### 4.2 Future work

1. **Integrate the biomimetic lung model**
2. **Re-identify the model and delay**
3. **Validate bilateral repeatability**

Extend the verified branch to independent left/right control across the required breathing range.